

Name:**Roll Number:********* WRITE BRIEF AND PRECISE ANSWERS. DO YOUR OWN WORK. DO NOT LOOK AROUND *********1. Briefly mention and describe the usage of each cloud model.**<https://learn.microsoft.com/training/modules/describe-cloud-compute/5-define-cloud-models>

Public cloud	Private cloud	Hybrid cloud
• No capital expenditures to scale up. • Applications can be quickly provisioned and deprovisioned. • Organizations pay only for what they use.	• Hardware must be purchased for start-up and maintenance. • Organizations have complete control over resources and security. • Organizations are responsible for hardware maintenance and updates.	• Provides the most flexibility. • Organizations determine where to run their applications. • Organizations control security, compliance, or legal requirements.

2. Describe the benefits of high availability and scalability in the cloud.<https://learn.microsoft.com/en-us/training/modules/describe-benefits-use-cloud-services/2-high-availability-scalability-cloud>

- When you're deploying an application, a service, or any IT resources, it's important the resources are available when needed. High availability focuses on ensuring maximum availability, regardless of disruptions or events that may occur.
- When you're architecting your solution, you'll need to account for service availability guarantees. Azure is a highly available cloud environment with uptime guarantees depending on the service. These guarantees are part of the service-level agreements (SLAs).

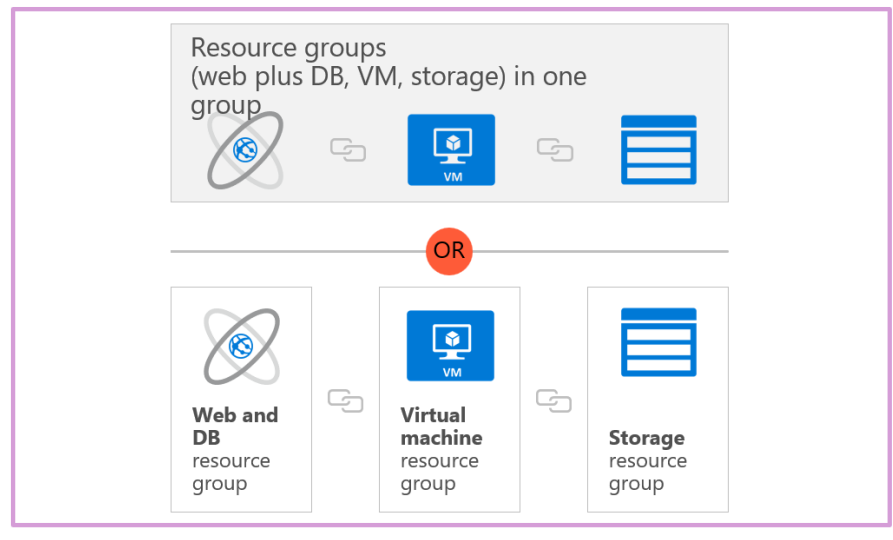
3. Briefly explain the purpose and usage of Azure Resource Groups<https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/6-describe-azure-management-infrastructure>

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A **resource group** is a container you use to manage and aggregate resources in a single unit.

- Resources can exist in only one resource group.
- Resources can exist in different regions.
- Resources can be moved to different resource groups.
- Applications can utilize multiple resource groups.



4. Brief explain virtual machine scale sets? How do you ensure same configuration of all the machines in a scale set?

<https://learn.microsoft.com/en-us/training/modules/describe-azure-compute-networking-services/2-virtual-machines>

- Virtual machine scale sets let you create and manage a group of identical, load-balanced VMs.
- The number of VM instances can automatically increase or decrease in response to demand, or you can set it to scale based on a defined schedule. Virtual machine scale sets also automatically deploy a load balancer to make sure that your resources are being used efficiently.

We can create an image from a VM which has all the software properly configured, and then use that image in VMSS to ensure that each VM is identically configured

5. Describe the following cloud benefits: a) security b) governance

<https://learn.microsoft.com/en-us/training/modules/describe-benefits-use-cloud-services/4-security-governance-cloud>

You can find a cloud solution that matches your security needs. If you want maximum control of security, infrastructure as a service provides you with physical resources but lets you manage the operating systems and installed software, including patches and maintenance. If you want patches and maintenance taken care of automatically, platform as a service or software as a service deployments may be the best cloud strategies for you.

Cloud features support governance and compliance. Things like set templates help ensure that all your deployed resources meet corporate standards and government regulatory requirements. Plus, you can update all your deployed resources to new standards as standards change. Cloud-based auditing helps flag any resource that's out of compliance with your corporate standards and provides mitigation strategies. Depending on your operating model, software patches and updates may also automatically be applied, which helps with both governance and security.

6. Describe consumption-based model and its benefits

<https://learn.microsoft.com/training/modules/describe-cloud-compute/6-describe-consumption-based-model>

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Cloud service providers operate on a consumption-based model, which means that end users only pay for the resources that they use.

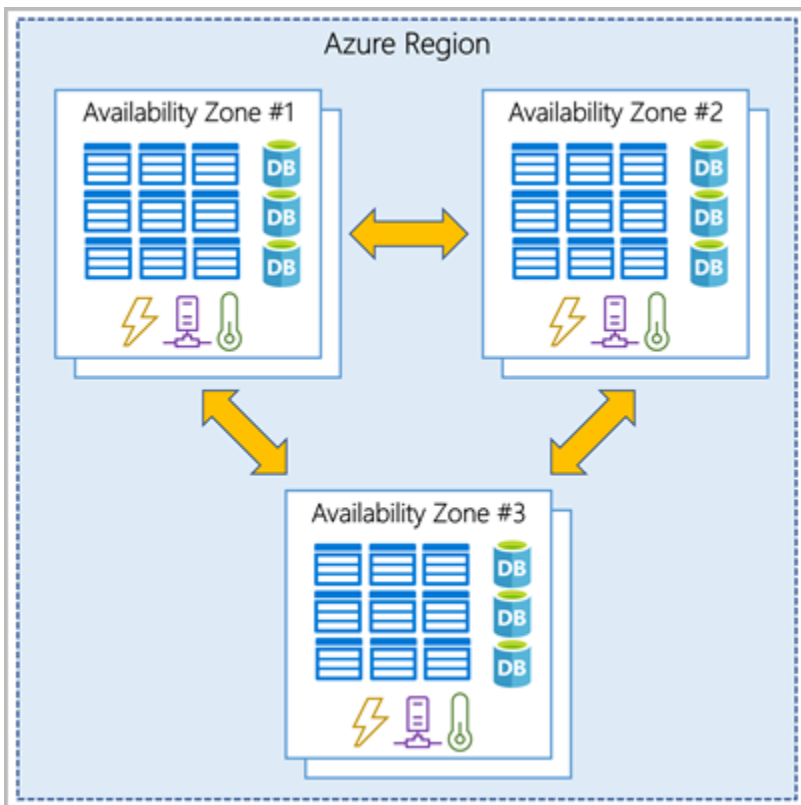
- Better cost prediction.
- Prices for individual resources and services are provided.
- Billing is based on actual usage.

7. Briefly describe the architecture and usage of availability zones

<https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/6-describe-azure-management-infrastructure>

Availability zones are physically separate datacenters within an Azure region. Each availability zone is made up of one or more datacenters equipped with independent power, cooling, and networking. An availability zone is set up to be an isolation boundary. If one zone goes down, the other continues working. Availability zones are connected through high-speed, private fiber-optic networks.

To ensure resiliency, a minimum of three separate availability zones are present in all availability zone-enabled regions. However, not all Azure Regions currently support availability zones.



8. Briefly compare the cloud services.

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IaaS	PaaS	SaaS
• The most flexible cloud service. • You configure and manage the hardware for your application.	• Focus on application development. • Platform management is handled by the cloud provider.	• Pay-as-you-go pricing model. • Users pay for the software they use on a subscription model.

9. Briefly explain the purpose of Azure Subscriptions

<https://learn.microsoft.com/en-us/training/modules/describe-core-architectural-components-of-azure/6-describe-azure-management-infrastructure>

In Azure, subscriptions are a unit of management, billing, and scale. Similar to how resource groups are a way to logically organize resources, subscriptions allow you to logically organize your resource groups and facilitate billing.

There are two types of subscription boundaries that you can use:

- **Billing boundary:** This subscription type determines how an Azure account is billed for using Azure. You can create multiple subscriptions for different types of billing requirements. Azure generates separate billing reports and invoices for each subscription so that you can organize and manage costs.
- **Access control boundary:** Azure applies access-management policies at the subscription level, and you can create separate subscriptions to reflect different organizational structures. An example is that within a business, you have different departments to which you apply distinct Azure subscription policies. This billing model allows you to manage and control access to the resources that users provision with specific subscriptions.